CreditLens

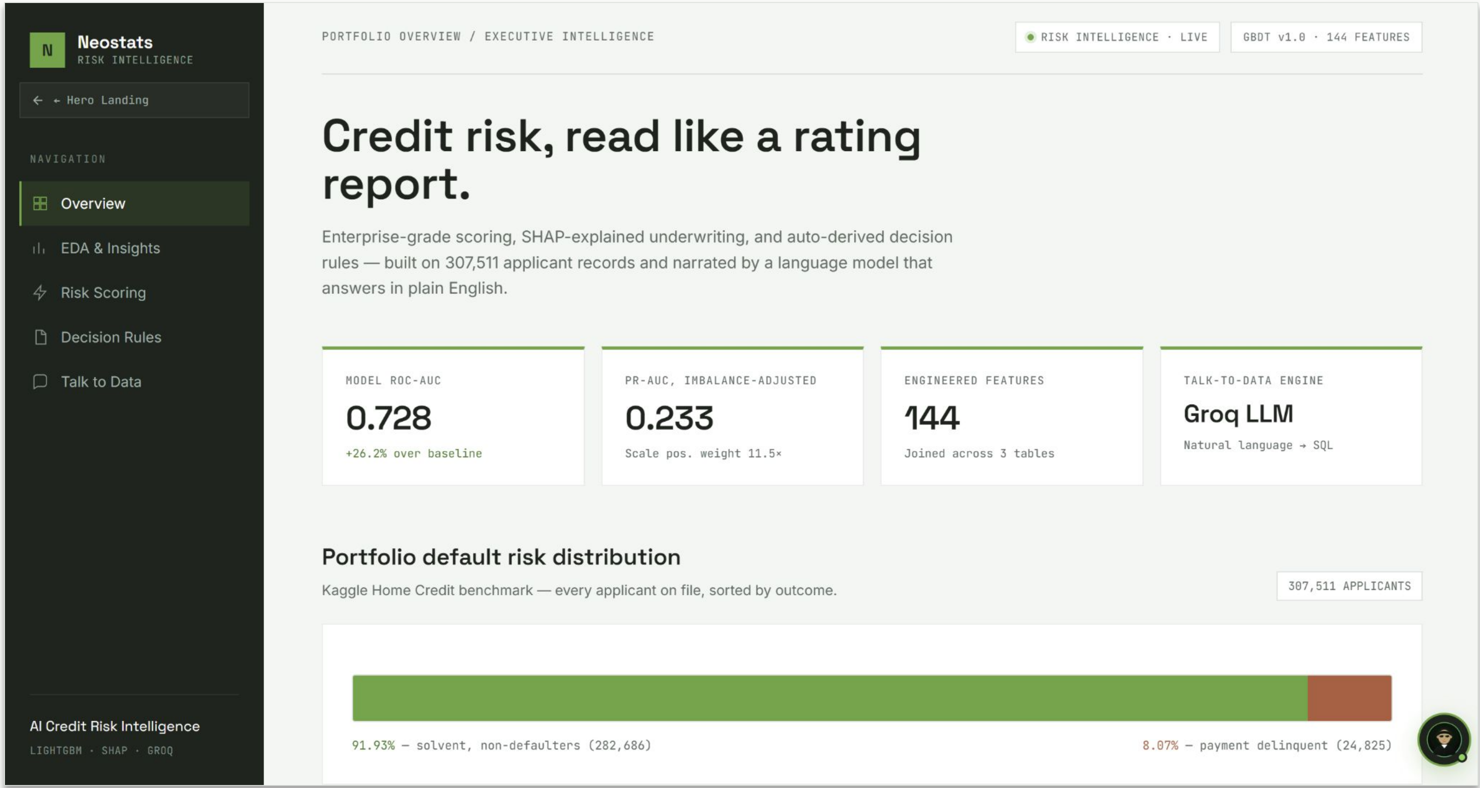
Explainable credit risk — from prediction to underwriting decision.

144	0.7285
ENGINEERED FEATURES	ROC-AUC

LightGBM + SHAP + Talk-to-Data

A working risk-intelligence interface built around the model.

PORTFOLIO OVERVIEW | EXPLAINABLE | ACTIONABLE



Default is rare. A wrong decision is not.

The portfolio is highly imbalanced, so model quality must be judged beyond accuracy.

8.07%
DEFAULT RATE

11.5 : 1
CLASS RATIO

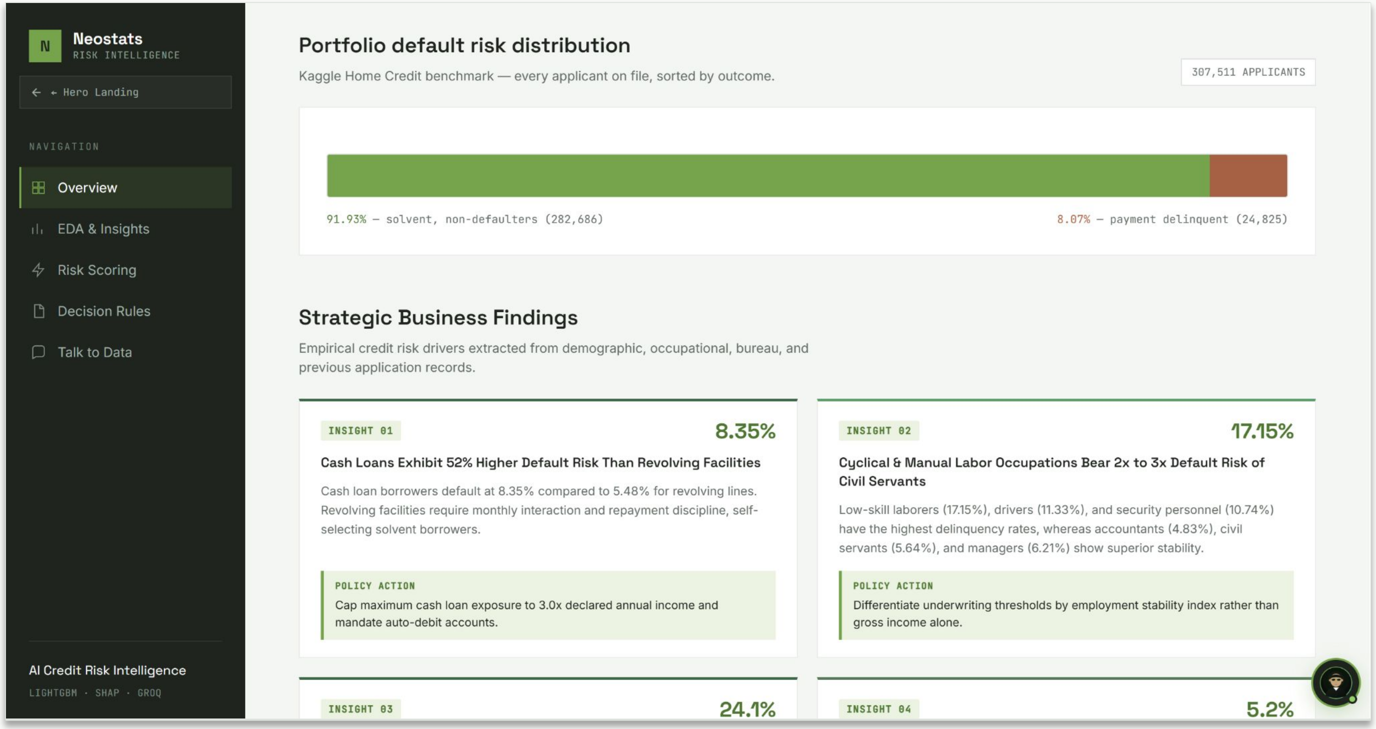
PR-AUC
PRIMARY SIGNAL

Why the imbalance matters

— **Cost-sensitive learning**
the minority class needs more weight.

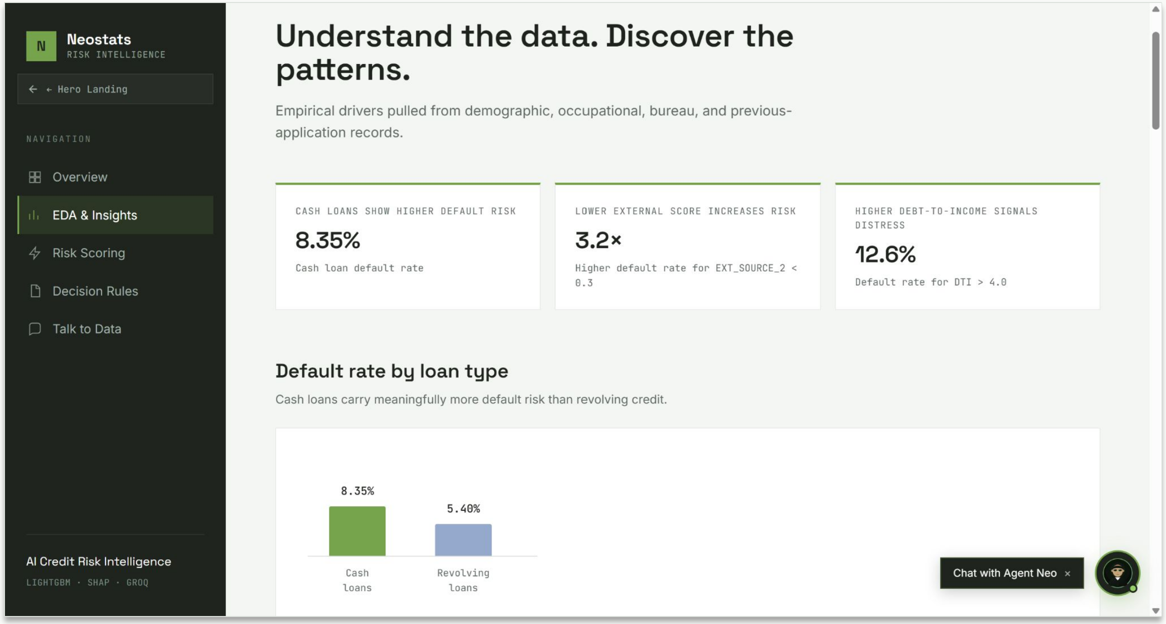
— **Stratified 5-fold CV**
keeps default cases represented in validation.

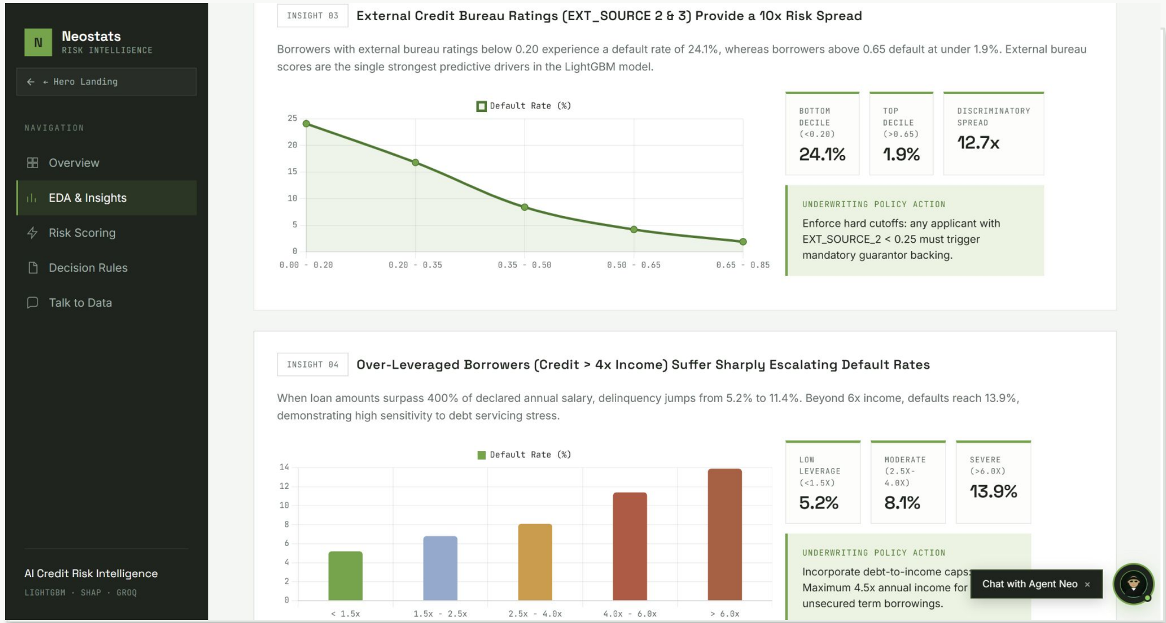
DECISION-FIRST



The raw portfolio becomes model-ready evidence.

EDA identifies the signals worth carrying into feature engineering and underwriting.



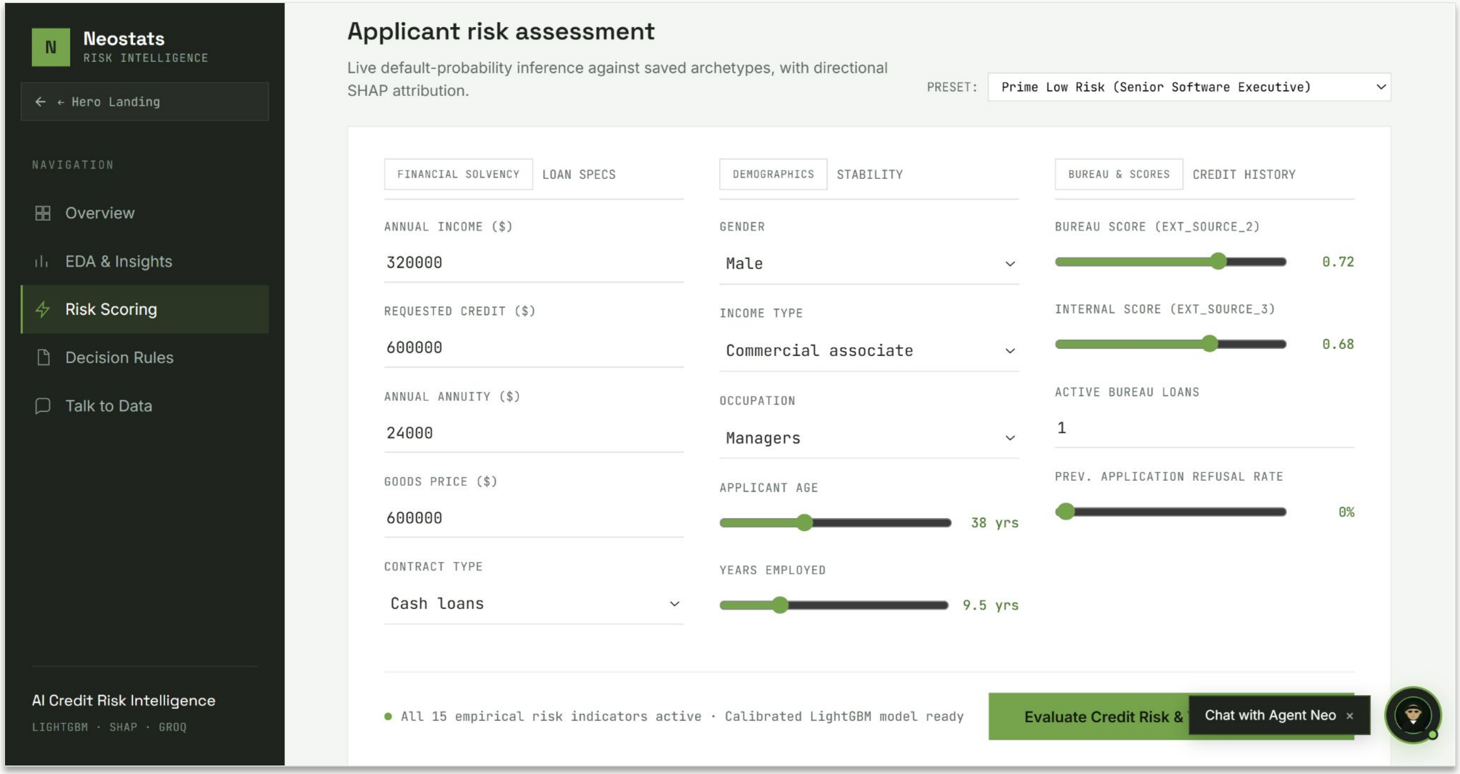


APPLICATION DATA → BUREAU + HISTORY → **FEATURE ENGINEERING** → **144 FEATURES**

Ratios • aggregates • external scores • historical behaviour

The score is only useful when it reaches a decision.

LightGBM ranks risk; the operating threshold converts probability into an underwriting action.



LIGHTGBM

800 estimators • max depth 7

0.7285
ROC-AUC

0.2330
PR-AUC

0.385
F1 THRESHOLD

<5 ms
INFERENCE TARGET

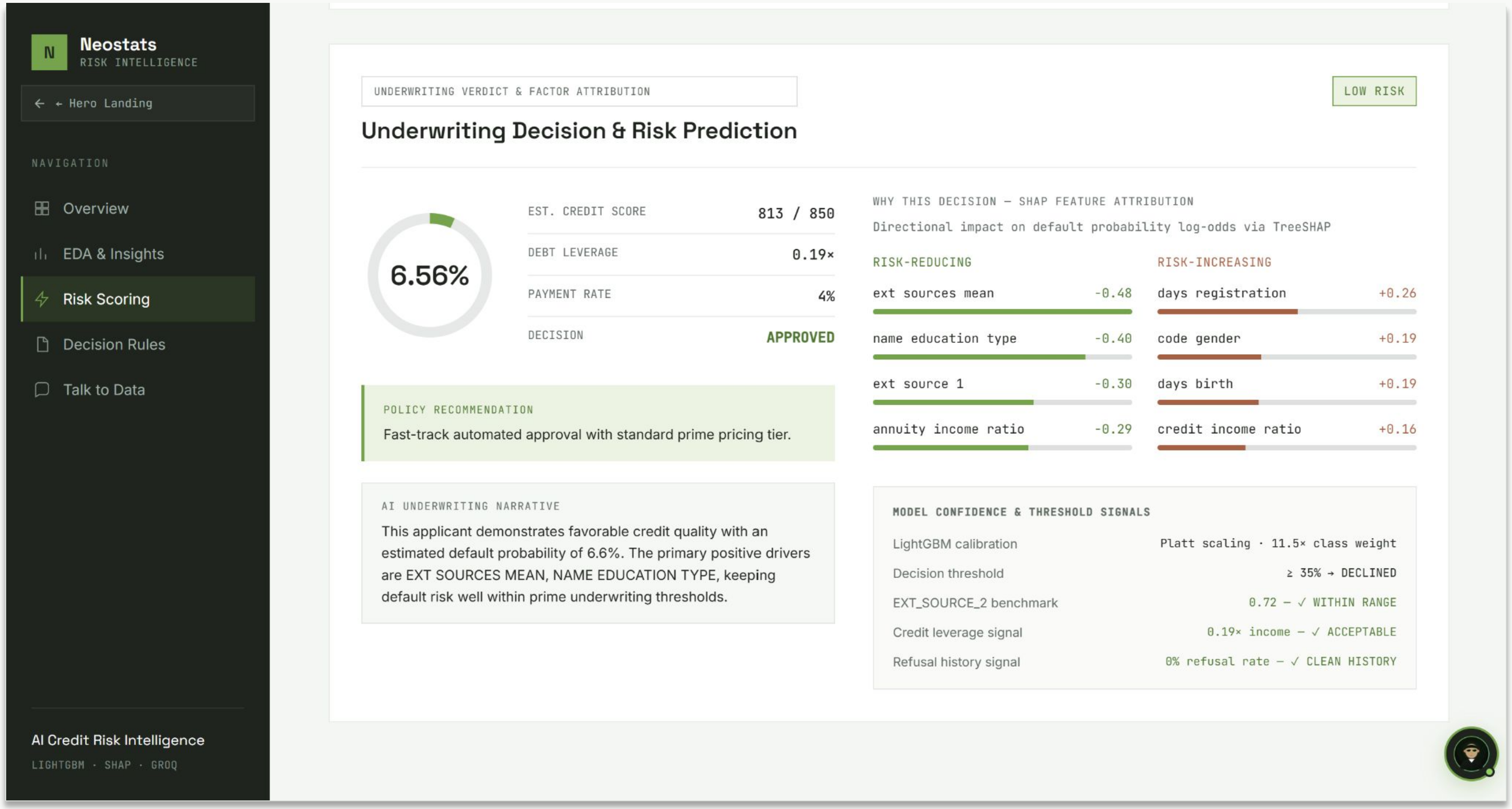
MODEL FLOW

features → score → threshold → decision

EARLY STOPPING

A prediction should be inspectable.

SHAP shows which applicant attributes pushed an individual risk score higher or lower.



WHAT MOVES RISK?

EXT_SOURCE_2 / 3

external score signals

DAYS_EMPLOYED

employment history

CREDIT_INCOME_RATIO

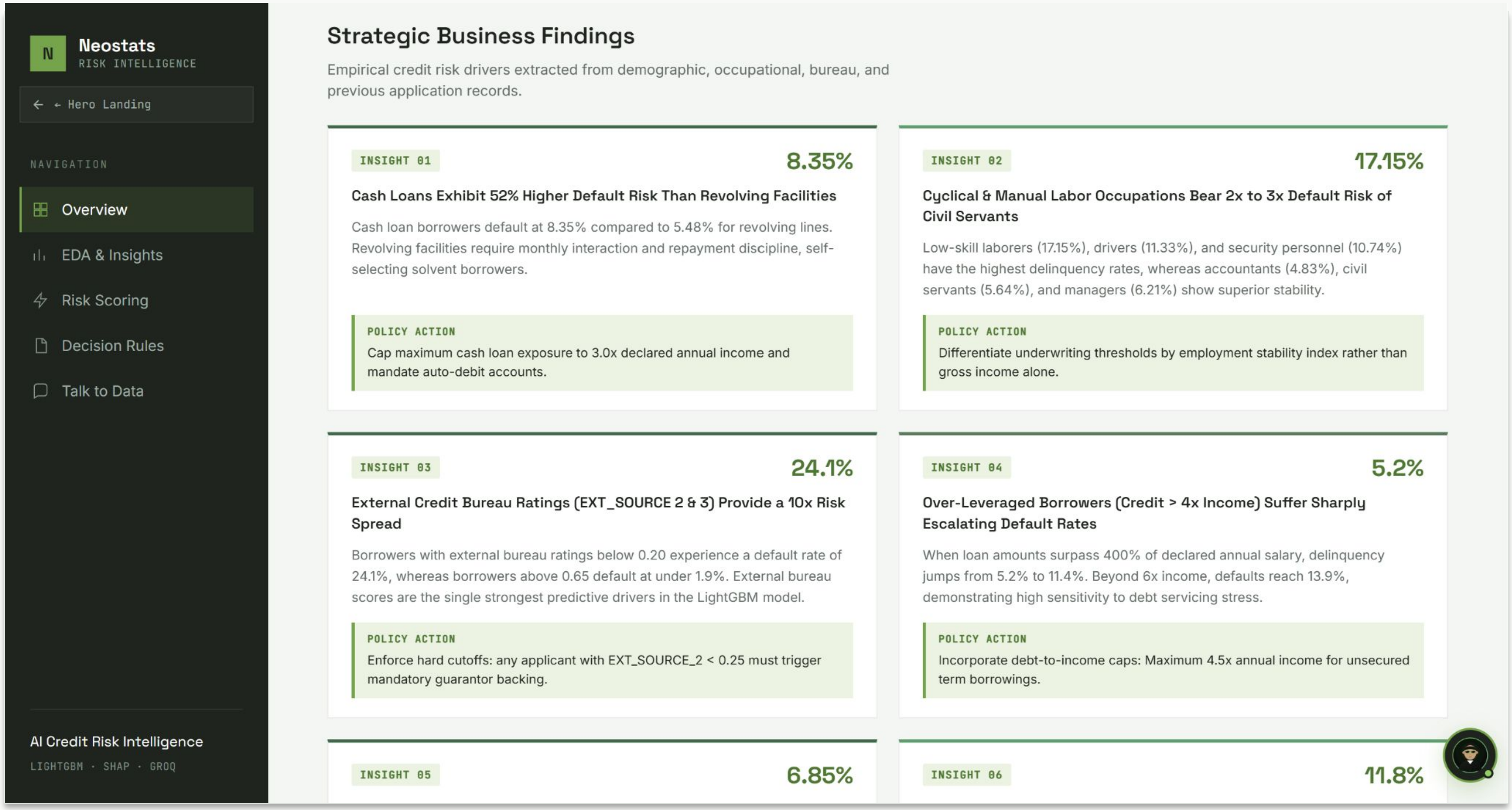
repayment pressure

LOCAL EXPLANATION

The model can explain one applicant — not only the portfolio average.

Risk signals become repeatable actions.

Seven policy rules translate model evidence into a consistent underwriting path.



DECISION TIERS

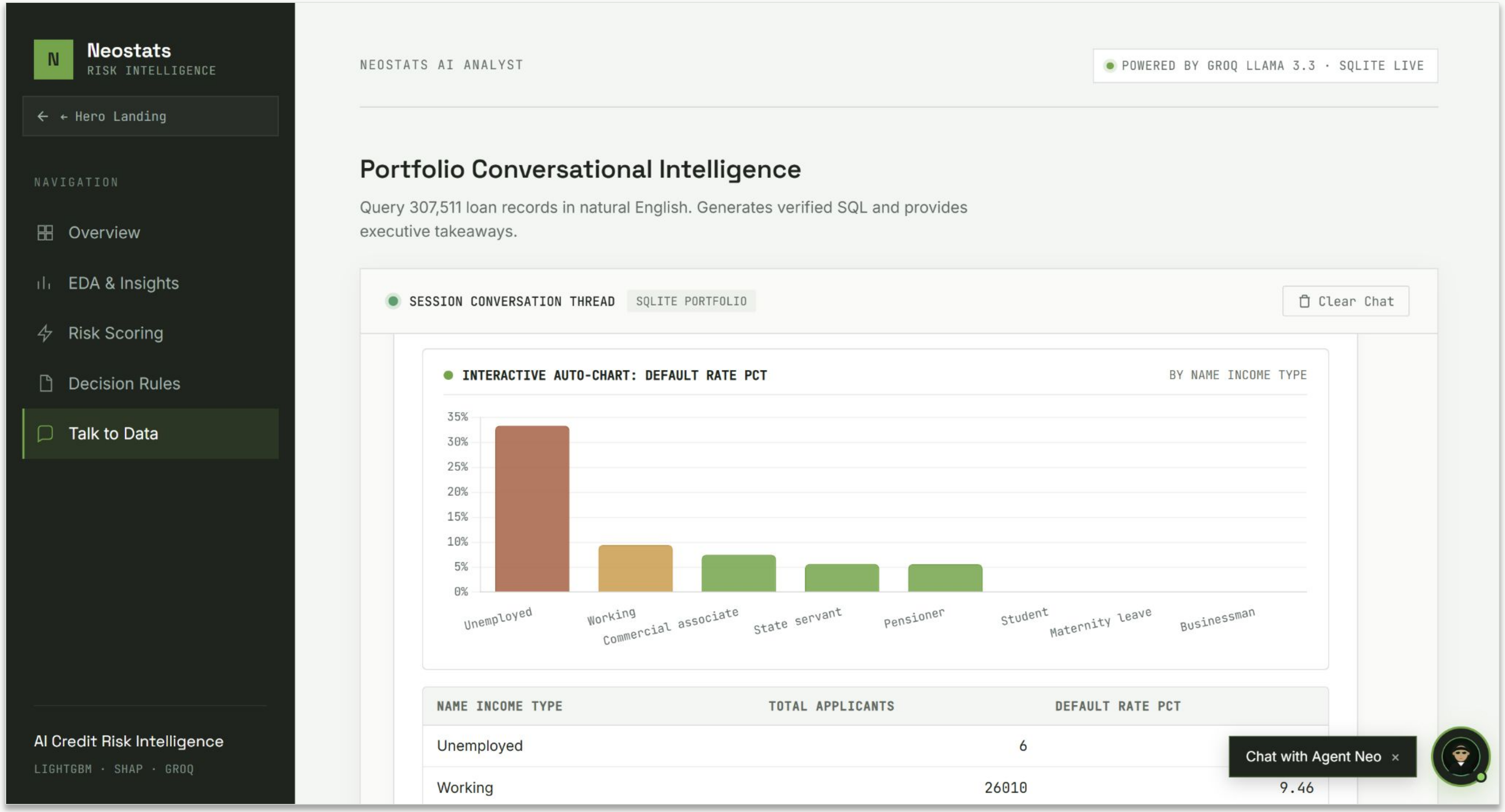
LOW	standard flow
MEDIUM	review / checks
HIGH	decline / escalate

7 RULES

Policy makes the model operational.

Ask the portfolio a question.

Natural language becomes constrained SQL, then returns a business-ready answer.



THE FLOW

01 ASK

02 GENERATE

03 VALIDATE

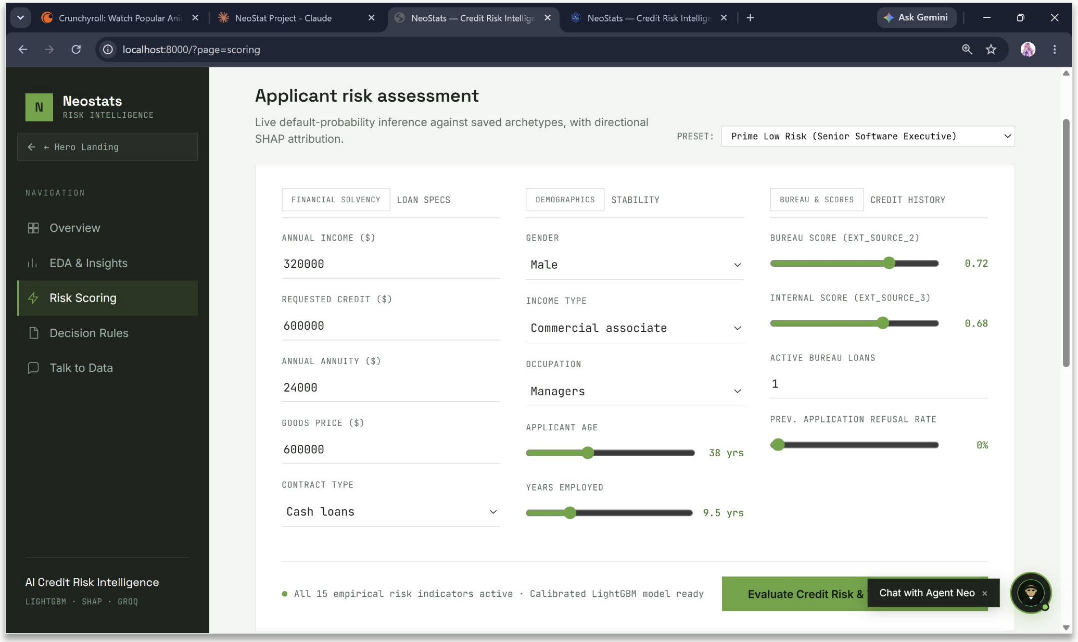
04 ANSWER

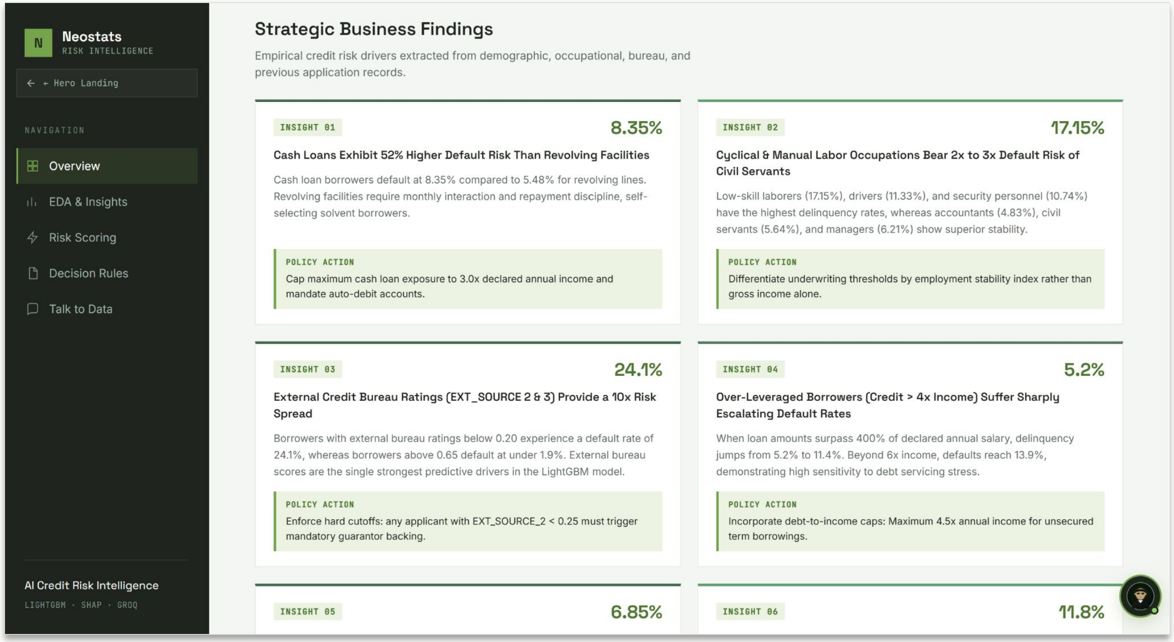
SELECT-only

mutation blocker • schema validator • auto-LIMIT

The finished product is more than one dashboard.

Different screens answer different questions — assessment, findings and conversational analysis.





APPLICANT RISK ASSESSMENT

Decision support

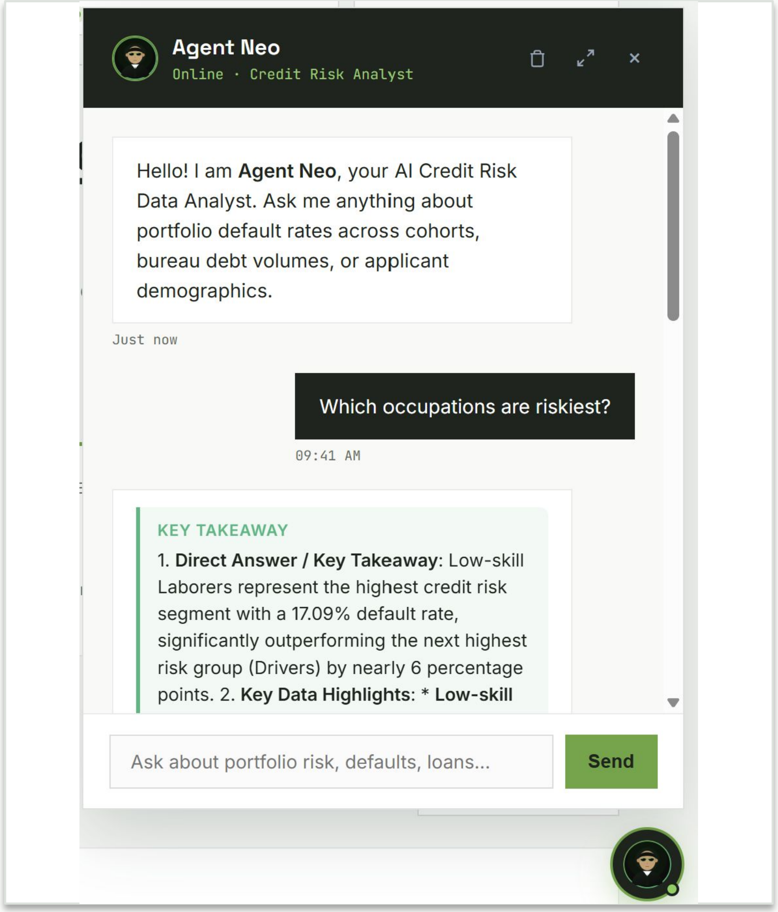
STRATEGIC BUSINESS FINDINGS

Portfolio interpretation

The product moves from an individual borrower to a portfolio-level view without leaving the same system.

A conversational layer on top of the risk system.

Agent Neo gives the user a natural-language entry point into portfolio intelligence.



WHAT THE AGENT ADDS

01 NATURAL LANGUAGE

Ask for insights without writing SQL.

02 DATA ACCESS

Connect questions to portfolio tables.

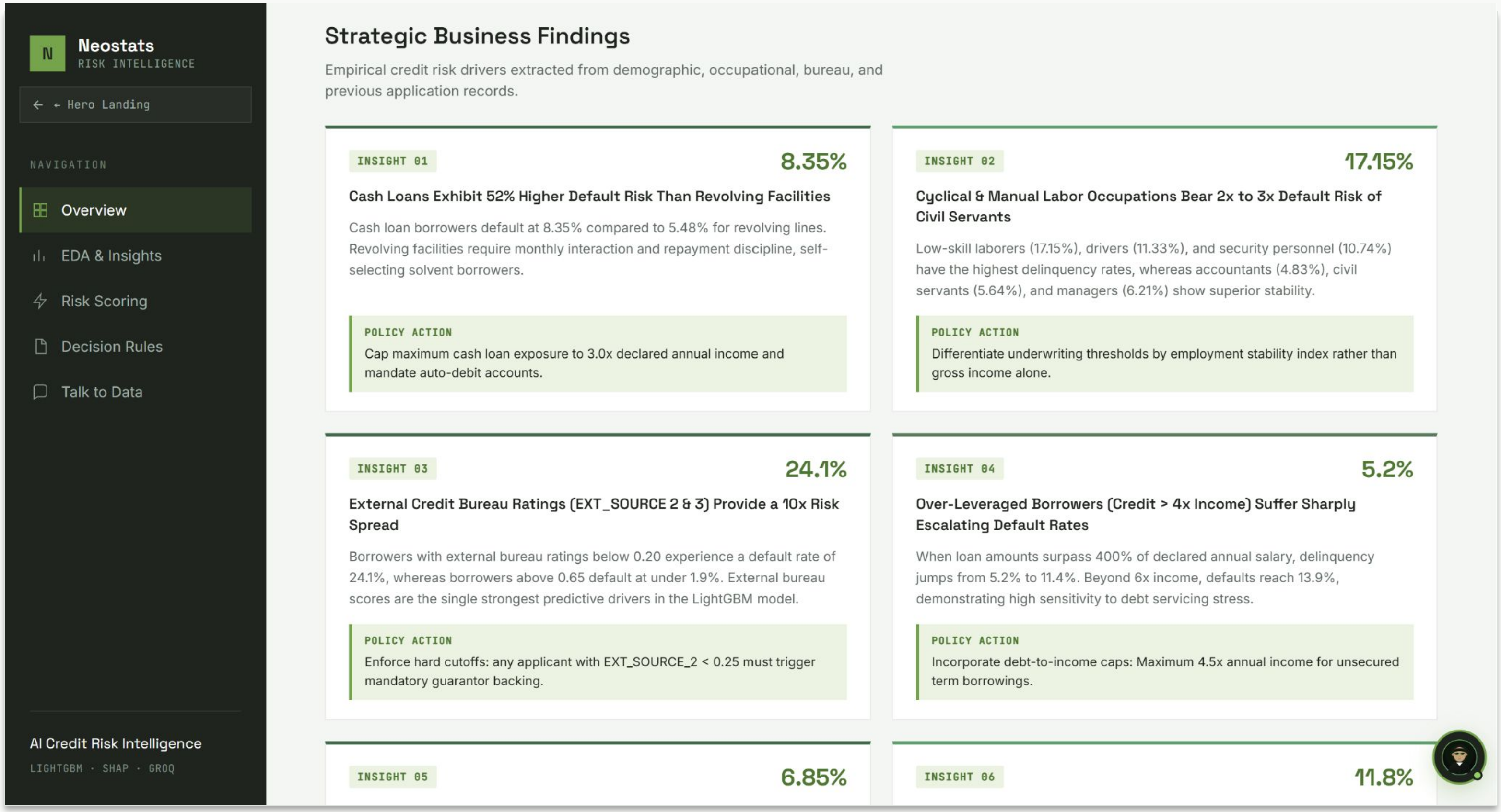
03 GUARDRAILS

Keep generated queries constrained.

GROQ | SELECT-ONLY | AUTO-LIMIT

The system turns model output into portfolio insight.

Strategic findings make the risk model useful beyond a single applicant.



FROM MODEL TO MANAGEMENT

— **Identify concentrations**

see where risk is accumulating.

— **Prioritize action**

focus review on material segments.

— **Explain the why**

connect findings back to model evidence.

PORTFOLIO LEVEL

A compact stack connects the intelligence end-to-end.

The system separates interface, API, model, data and language services.

USER EXPERIENCE	web app + Agent Neo
API	FastAPI REST services
INTELLIGENCE	LightGBM + SHAP
DATA	SQLite portfolio data
LANGUAGE	Groq-assisted Talk-to-Data

Simple enough to understand. Modular enough to extend.

Built to run, not just demo.

Reliability comes from reproducible deployment and disciplined delivery.

01 **CONTAINERIZED**

Dockerized runtime

02 **OBSERVABLE**

health checks + visibility

03 **VERSIONED**

Git-based delivery

04 **TESTED**

regression safety

05 **CONTROLLED**

uv + requirements

06 **MODULAR**

ready for new models

DELIVERY

reproducible • observable • testable • maintainable

One workflow replaces disconnected risk tasks.

The value is the connection between model evidence and the action that follows.

01 **SCORE**
rank applicant risk

→

02 **EXPLAIN**
show the drivers

→

03 **POLICY**
apply repeatable rules

→

04 **ANALYZE**
query the portfolio

WHAT THE USER GETS

- **Faster decisions**
  less switching between tools and views.
- **More transparent decisions**
  model output comes with an explanation.
- **Portfolio awareness**
  individual decisions connect to broader risk patterns.

Explainable credit risk, built for action.

Prediction is only the beginning.
The system connects score → explanation → policy → portfolio insight.

LIGHTGBM | SHAP | POLICY | TALK-TO-DATA

Next: retraining • fairness auditing • streaming intelligence

CORE WORKFLOW

SCORE
↓
EXPLAIN
↓
APPLY POLICY
↓
ASK THE PORTFOLIO